



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY

A.S Nagar, Elupatti – Thanjavur.

**SKILL
WALLET**
A SmartBridge Product



Team ID : SWTID-2026-8294

Team Size : 4

Team Leader : Muthukumar K

Team member : Manoj L

Team member : Prajin Oaswald A

Team member : Kamalesh P

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

Project Report

Table of Contents

1. INTRODUCTION	4
User Story	4
1.1 Project Overview	4
1.2 Purpose	4
1.3 Scope of the Project	4
1.4 Industry Background	4
1.5 Key Concepts Used in this Project	5
2. IDEATION PHASE	6
2.1 Problem Statement	6
2.2 Empathy Map Canvas	7
2.3 Brainstorming and Idea Prioritization	8
3. REQUIREMENT ANALYSIS PHASE	9
3.1 Customer Journey Map	9
3.2 Solution Requirements	9
3.3 Data Flow Diagram	11
3.4 Technology Stack	11
4. PROJECT DESIGN PHASE	12
4.1 Problem-Solution Fit	12
4.2 Proposed Solution	12
4.3 Solution Architecture	12
5. PROJECT PLANNING AND SCHEDULING	14
5.1 Project Planning	14
6. PROJECT DEVELOPMENT PHASE	16
What was learned and used	16
Milestone 1: Salesforce Developer Account	16
Milestone 2: Data Management (Objects)	17
Milestone 3: Record-Triggered Flow (set priority)	21
Milestone 4: Autolaunched Flow (Support Ticket Priority Flow)	24
Milestone 5: Agentforce Configuration	27
7. FUNCTIONAL AND PERFORMANCE TESTING	31
7.1 Test cases	31
7.2 Test evidence	31
7.3 Performance	33

7.4 Observations	33
7.5 Issues Found and Fixed	33
8. ADVANTAGES AND DISADVANTAGES	35
Advantages	35
Disadvantages	35
9. CONCLUSION	36
10. FUTURE SCOPE	36
11. APPENDIX	37
A. Custom object	37
B. Key Salesforce records	37
C. Flows	37
D. Agentforce	37
E. Source files	37
12. REFERENCES	38

1. INTRODUCTION

User Story

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to look at the details of a customer support ticket, decide its priority (Critical, High, Medium or Low as configured in each flow), and fill in the supporting values that the support team needs: a priority score and the team that should handle the ticket. The solution uses a custom object named **support ticket intelligence**, a record-triggered Flow that works as soon as a ticket is saved, an Autolaunched Flow that can look up a customer's ticket by Account Name, and an Agentforce subagent named **Support Ticket Priority Analysis** that is connected to that Autolaunched Flow.

1.1 Project Overview

Customer support teams receive a large number of tickets every day. Deciding which ticket is urgent and who should handle it is often done by reading each ticket by hand. This takes time, differs from one person to another, and lets urgent problems wait. This project builds a Salesforce based solution that reads the ticket information, classifies how urgent the ticket is, gives it a score, chooses a support team, and lets a user ask Agentforce for the priority of a customer's ticket.

1.2 Purpose

- Classify support tickets automatically as the ticket is created.
- Give each ticket a numeric priority score so tickets can be sorted.
- Fill the assigned support team automatically (Billing Support, Account Support or Technical Support).
- Retrieve the support ticket linked to a customer Account and classify it as High, Medium or Low through an Autolaunched Flow.
- Use an Agentforce subagent and action to start that Flow from a conversation.
- Reduce manual prioritization effort and improve response time.

1.3 Scope of the Project

The project is developed and tested in a Salesforce Developer Edition org. It covers the data model, two Flows, one Agentforce agent with one subagent and one flow-based action, and the testing of each part. Priority prediction is **rule based**: it uses conditions on ticket fields and keywords inside Flow. It does not use a trained machine learning model. Features such as queues, automatic task creation and SLA monitoring are listed under Future Scope.

1.4 Industry Background

Salesforce's Seventh Edition State of Service report is based on a survey of 6,500 service professionals worldwide. It reports that AI resolved about 30 percent of service cases in 2025 and is expected to resolve about 50 percent by 2027 [1][3]. Respondents also expect AI agents to lower service costs and case resolution times by about 20 percent on average, and 88 percent said conversational AI speeds up resolution [2].

These figures come from a vendor survey, so they show the direction of the industry and are not a guarantee of results. They support the idea behind this project: support teams are moving routine

sorting and lookup work to automation and AI, so that people can spend their time on the tickets that need them. Ticket priority and routing are among the first steps that can be automated.

1.5 Key Concepts Used in this Project

Concept	Meaning in this project
Custom object	A table defined in Salesforce to store data. Here: support ticket intelligence.
Record-triggered Flow	A Flow that runs automatically when a record is created or changed. Here: set priority.
Autolaunched Flow	A Flow with no trigger that another tool starts. An agent action of type Flow is backed by an autolaunched Flow [4][8]. Here: Support Ticket Priority Flow.
Agent	Uses a large language model and a reasoning engine to decide when to launch subagents and actions during a conversation [6].
Subagent	Groups one job of the agent: its description, scope, reasoning instructions and the actions available for that job [5][8]. Here: Support Ticket Priority Analysis.
Agent action	A capability the agent can run, such as a Flow. The reasoning engine reviews the action names, descriptions and inputs, so clear descriptions matter [6][7].
Reasoning instructions	Plain language rules that tell the agent how and when to use its actions [8].

2. IDEATION PHASE

2.1 Problem Statement

The customer problem statement template below lists the people affected, what they are trying to do, what stops them, and how it makes them feel.

	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	prioritization is done manually	Frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind routine ones	support teams handle large ticket volumes	Concerned
PS-3	Support Manager	assign tickets to the right support team	assignment needs manual decisions	urgency is not analyzed in a consistent way	Overwhelmed
PS-4	Support Agent	spot urgent issues from the description	important keywords can be missed	descriptions contain many different urgency signals	Uncertain
PS-5	Support Manager	find the latest ticket of a customer quickly	the ticket must be searched for by hand	tickets are spread across many accounts	Concerned

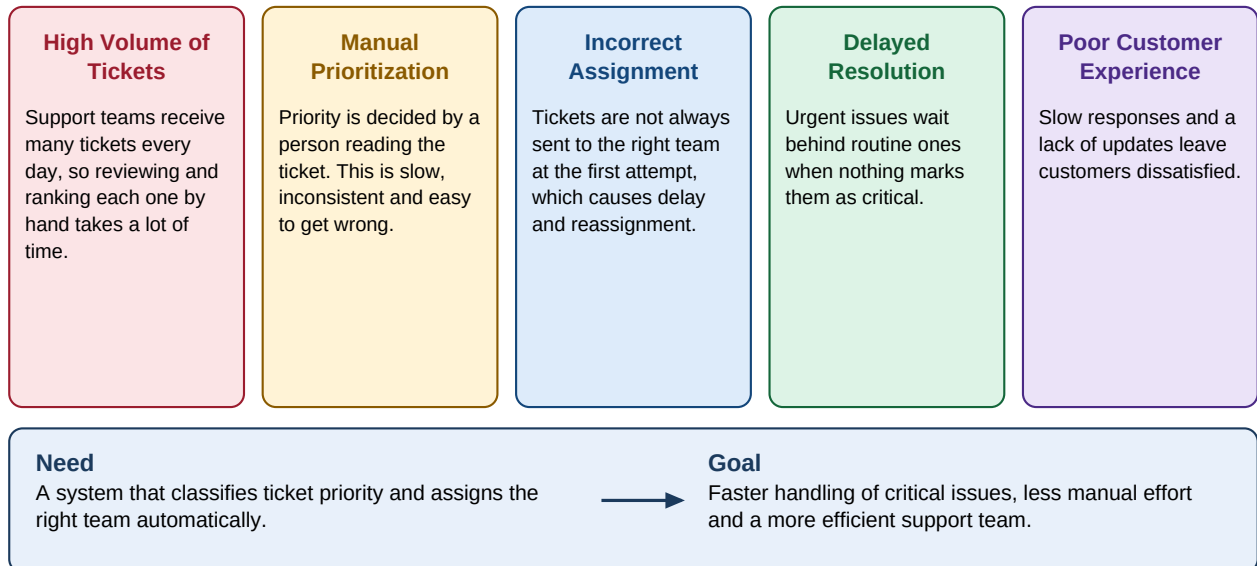


Figure 2.1: Problem statements in customer support

2.2 Empathy Map Canvas

User persona: Support Agent. The empathy map below shows what the agent says, thinks, sees and hears, together with the pains the project removes and the gains it delivers.

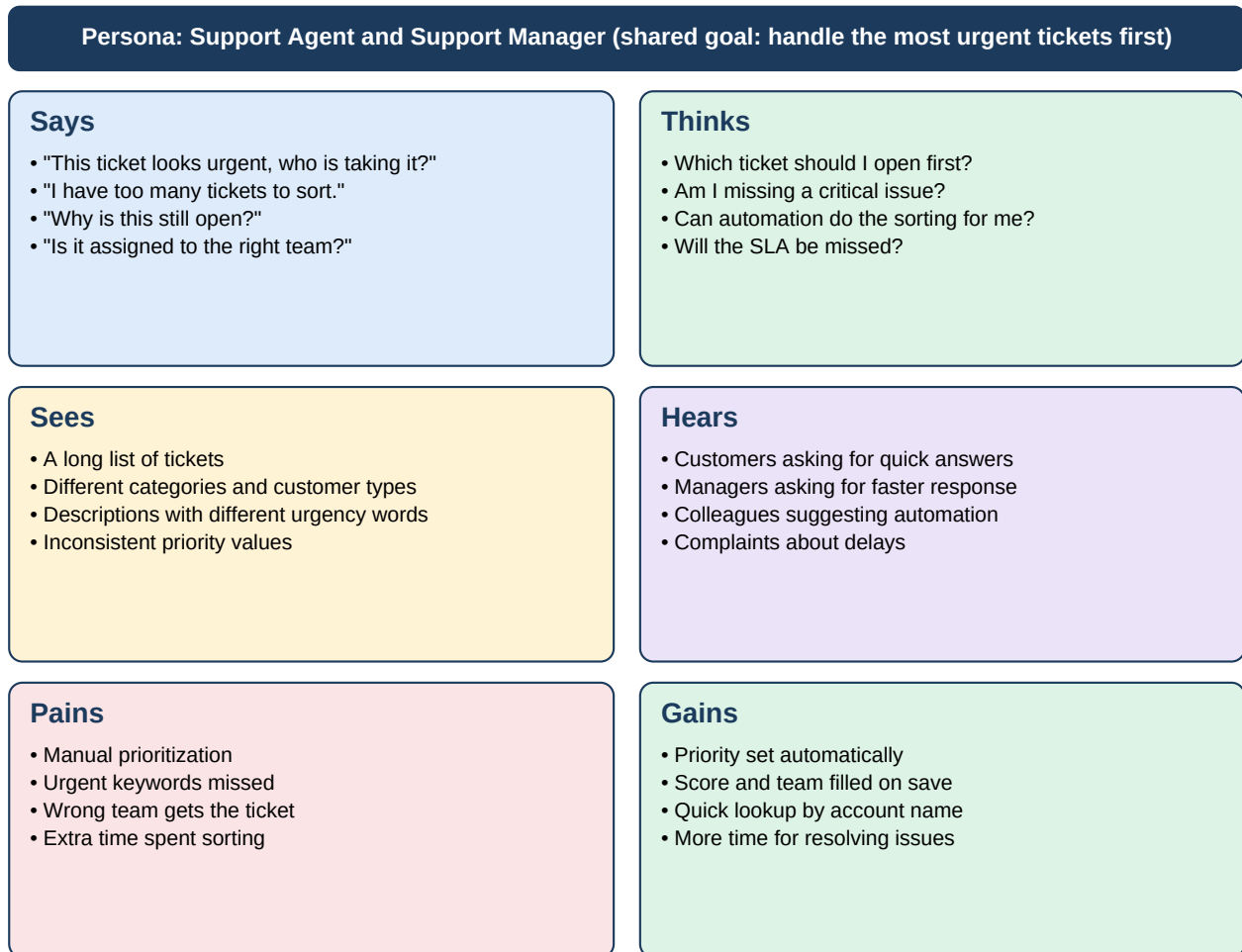


Figure 2.2: Empathy map canvas

2.3 Brainstorming and Idea Prioritization

Ideas considered

- Manual ticket registers
- Spreadsheet based ticket tracking
- A standalone support portal
- Salesforce CRM with Flow automation and Agentforce

Ideas were then compared on impact and feasibility. Ideas with high impact that can be built with clicks inside Salesforce were chosen.

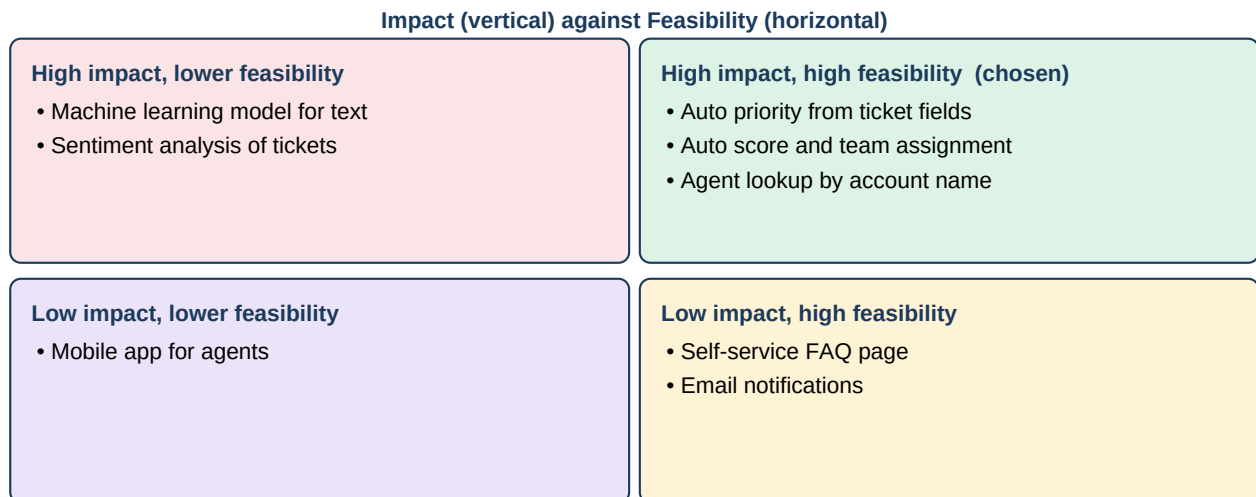


Figure 2.3: Idea prioritization matrix

Final selection

Salesforce CRM with Flow and Agentforce was selected because it gives:

- Automation through Salesforce Flow
- Conversational access through Agentforce
- Automatic priority classification and scoring
- Automatic team assignment
- Reduced manual support effort

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The journey map follows a ticket from the moment a customer reports a problem until it is resolved.

Stage	Customer / user action	Touchpoint	System action	Outcome
1. Raise ticket	Reports the problem	Support portal or agent	Ticket record created in support ticket intelligence	Ticket logged
2. Auto analysis	No action needed	Record-triggered Flow	Formulas read subject, category and customer type	Priority identified
3. Score and team	No action needed	Record-triggered Flow	Assignment writes Predict Priority, Priority Score and Assigned Team	Ticket routed
4. Ask the agent	Gives the Account Name	Agentforce conversation	Subagent asks for the account and starts the action	Request understood
5. Lookup	Waits	Autolaunched Flow	Get Account and Get Ticket find the linked ticket	Ticket found
6. Classify	Waits	Autolaunched Flow	Decision checks the description for urgency words	High, Medium or Low
7. Respond	Reads the reply	Agentforce conversation	Record ID, priority level and message returned	User informed
8. Resolve	Agent works on the ticket	Salesforce	Status updated by the support team	Ticket closed

Figure 3.1: Customer journey map

3.2 Solution Requirements

Functional requirements

FR No	Requirement	Met by
FR-1	Create and manage support tickets	Custom object support ticket intelligence
FR-2	Link each ticket to a customer Account	Customer lookup field to Account
FR-3	Analyze ticket details	Formula resources and Decision element
FR-4	Classify priority automatically	Flow: set priority and Support Ticket Priority Flow
FR-5	Give a priority score	Priority Score field filled by the Flow
FR-6	Assign the support team	Assigned Team field filled by the Flow
FR-7	Look up a customer's ticket by Account Name	Autolaunched Flow with Get Records
FR-8	Conversational access	Agentforce subagent and flow action

Non-functional requirements

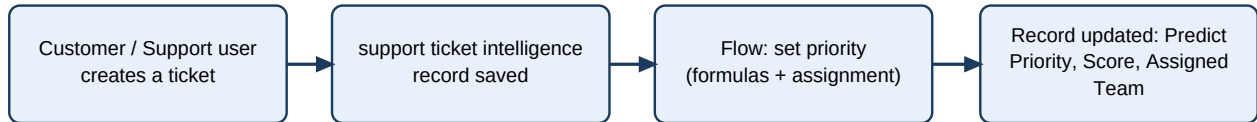
NFR	Description
Usability	Clear Agentforce interaction and understandable priority messages.
Reliability	The same ticket data always produces the same priority because the rules are fixed in Flow.

NFR	Description
Performance	The flows finish in about one to two seconds (Flow Debug showed 1.25 seconds for one run).
Security	Access to ticket and account records is controlled by Salesforce permissions.
Maintainability	Keywords, scores and team rules can be changed in the Flow without code.
Scalability	Salesforce automation handles growing ticket volumes.

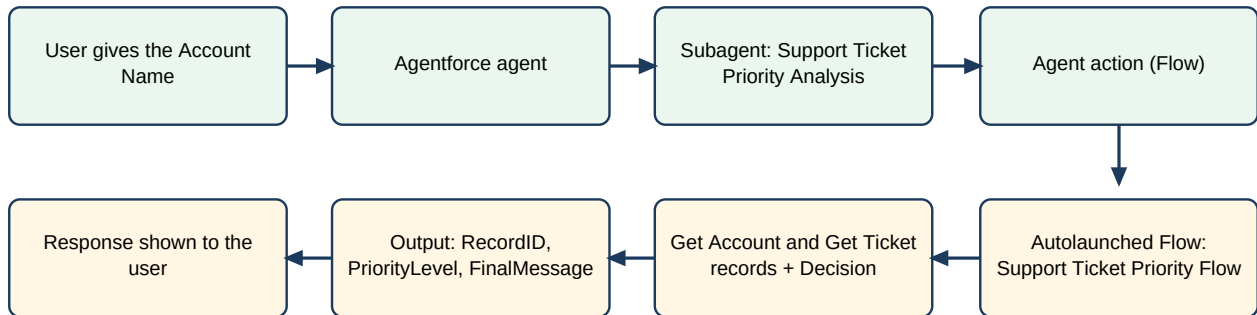
3.3 Data Flow Diagram

The diagram shows the two flows of data in the system. Flow A runs inside Salesforce when a ticket is saved. Flow B starts when a user asks the Agentforce agent for the priority of a customer's ticket.

Flow A: ticket creation (record-triggered)



Flow B: priority lookup through Agentforce



Data stores used: Account object and support ticket intelligence custom object

Figure 3.3: Data flow diagram

3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
Database	support ticket intelligence custom object (API name support_ticket__c) with the standard Account object
Logic	Record-triggered Flow (set priority) and Autolaunched Flow (Support Ticket Priority Flow)
AI	Agentforce Employee Agent
Agent configuration	Subagent Support Ticket Priority Analysis with a flow-based agent action
Automation elements	Formula resources, Decision, Assignment, Get Records
Output	Field values on the ticket record and an Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Record-triggered Flow sets Predict Priority when a ticket is created	Faster identification of urgent cases
Urgent issues may be missed	Keywords such as urgent, crash, down and payment failed mark a ticket Critical	Critical cases get attention first
No way to sort tickets	Flow writes a Priority Score (90, 70, 50 or 20)	Tickets can be sorted and filtered
Manual team assignment	Flow fills the Assigned Team from the ticket category	More consistent routing
Finding a customer's ticket takes time	Autolaunched Flow finds the ticket from the Account Name	Quick lookup
Need for conversational access	Agentforce subagent starts the flow through an action	AI assisted interaction

Figure 4.1: Problem-solution fit

4.2 Proposed Solution

Parameter	Description
Problem	Manual ticket prioritization and assignment delays the handling of critical customer support issues.
Solution	A Salesforce based system that predicts ticket priority, scores it, assigns a support team, and answers priority questions through Agentforce.
Innovation	Rule based Flow automation combined with an Agentforce subagent that calls an Autolaunched Flow from a conversation.
Impact	Faster identification of critical tickets, less manual effort and better use of the support team's time.
Business model	Customer support operations with customers, support agents, support managers and AI assisted ticket handling.
Scalability	More categories, keywords, teams and priority levels can be added by editing the Flow.

4.3 Solution Architecture

Components: Users, Agentforce agent and subagent, the agent action, the two Flows, and the data objects.

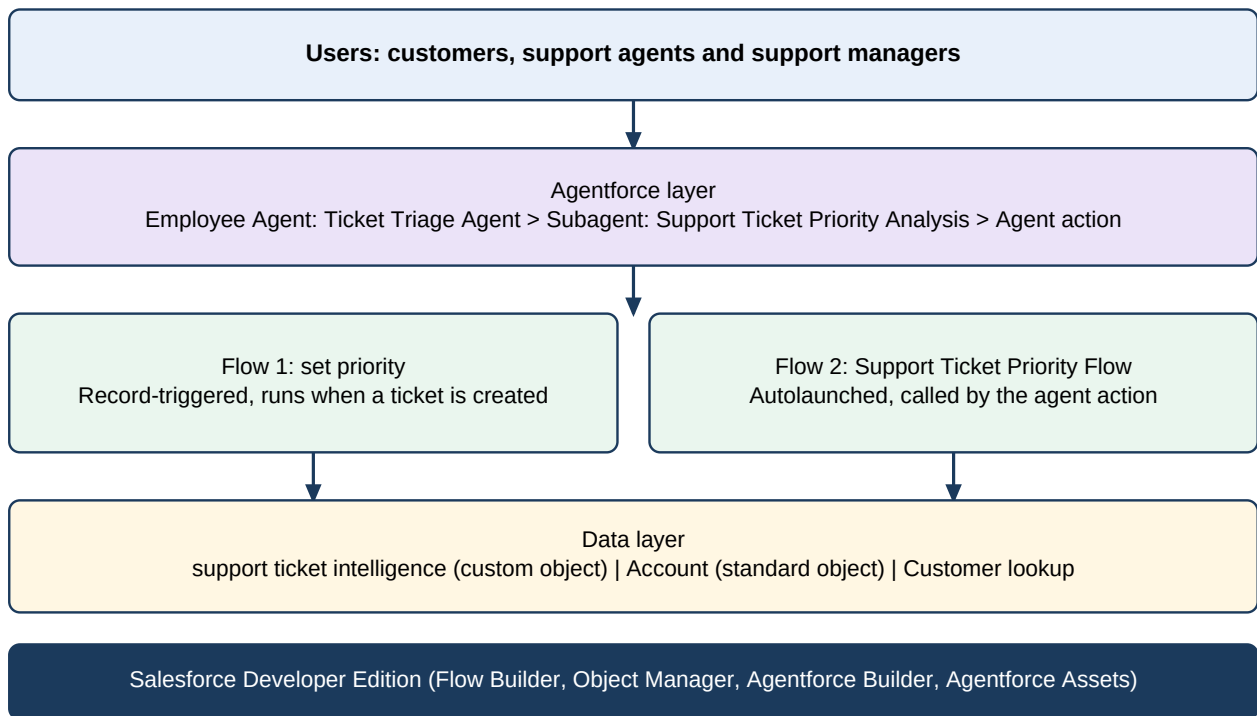


Figure 4.3: Solution architecture

5. PROJECT PLANNING AND SCHEDULING

5.1 Project Planning

- Agile methodology with a short sprint for each working day of the project sessions.
- Epics are broken into user stories, and each story has story points.
- Velocity is the number of story points completed in a sprint.

Product backlog, sprint schedule and estimation

Sprint	Epic	Story no.	User story / task	Points	Priority	Team member
Sprint-1	Developer setup	USN-1	As a system administrator, I want a Salesforce Developer Edition org so that I can build the support ticket solution.	3	High	K Muthukumar
Sprint-2	Data modeling	USN-2	As an admin, I want the support ticket intelligence object with the required fields so that ticket data is stored in an organized way.	5	High	K Muthukumar
Sprint-2	Data modeling	USN-3	As a support user, I want each ticket linked to an Account so that a customer's ticket can be found.	3	High	K Muthukumar
Sprint-3	Automation	USN-4	As a support agent, I want a record-triggered Flow to set Predict Priority, Priority Score and Assigned Team when a ticket is created.	8	High	K Muthukumar
Sprint-3	Automation	USN-5	As a support agent, I want an Autolaunched Flow that finds a customer's ticket from the Account Name and classifies its priority.	8	High	K Muthukumar
Sprint-4	Agentforce	USN-6	As a support user, I want an Agentforce subagent connected to the Flow so that I can ask for ticket priority in a conversation.	8	High	K Muthukumar
Sprint-4	Testing and documentation	USN-7	As a project owner, I want the solution tested and documented so that it can be submitted.	5	Medium	K Muthukumar

Project tracker, velocity and burndown chart

Sprint	Total story points	Duration	Sprint start	Planned end	Points completed	Release (actual)
Sprint-1	3	1 day	Day 1	Day 1	3	Day 1
Sprint-2	8	1 day	Day 2	Day 2	8	Day 2
Sprint-3	16	1 day	Day 3	Day 3	16	Day 3
Sprint-4	13	1 day	Day 4	Day 4	13	Day 4

Total story points are 40 over four sprints, so the average velocity is 10 points per sprint (40 divided by 4).

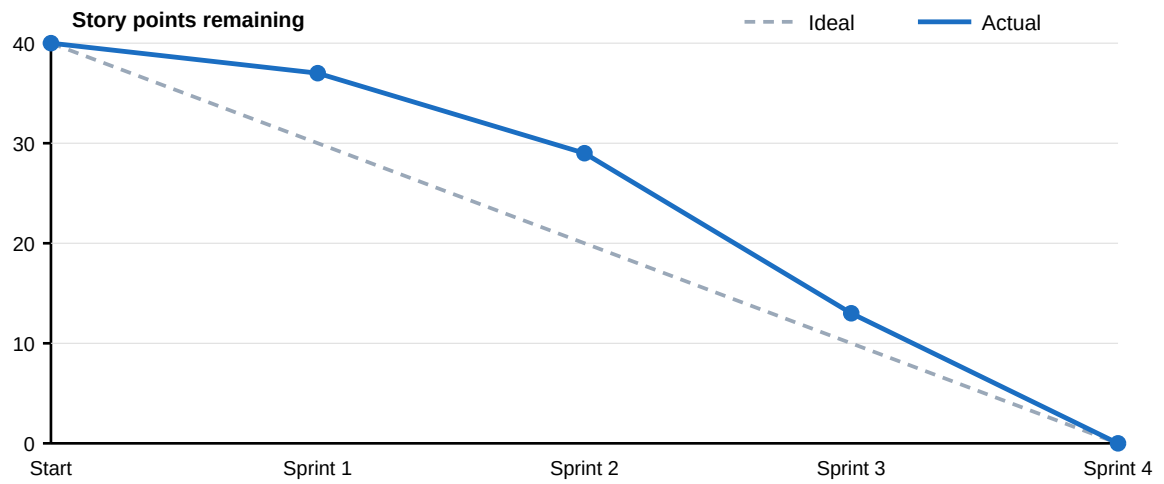


Figure 5.1: Sprint burndown chart

6. PROJECT DEVELOPMENT PHASE

What was learned and used

- Data modeling with custom objects
- Fields and relationships (picklists, lookup, auto number)
- Record-triggered Flows
- Autolaunched Flows with input and output variables
- Agentforce Builder: agents, subagents and agent actions

Milestone 1: Salesforce Developer Account

Activity 1: Creating the developer org

1. Open the Salesforce Developer Edition sign-up page.
2. Fill in first name, last name, email, role (Developer), company (college name), country (India), postal code and a unique username.
3. Select Sign me up.

Activity 2: Account activation

1. Open the verification email and select Verify Account.
2. Set a password and answer the security question.
3. Salesforce opens the Setup home page of the new org.

Milestone 2: Data Management (Objects)

Activity 1: Create the custom object

1. In Setup open Object Manager, then select Create and Custom Object.
2. Label: **support ticket intelligence**. Plural label: support ticket intelligences.
3. Record Name: Ticket Number. Data type: Auto Number. Display format: TKT-{0000}. Starting number: 1.
4. Tick Allow Reports, Allow Activities and Allow Search.
5. Set Deployment Status to Deployed and tick Launch New Custom Tab Wizard after saving.
6. Select Save, then complete the tab wizard.

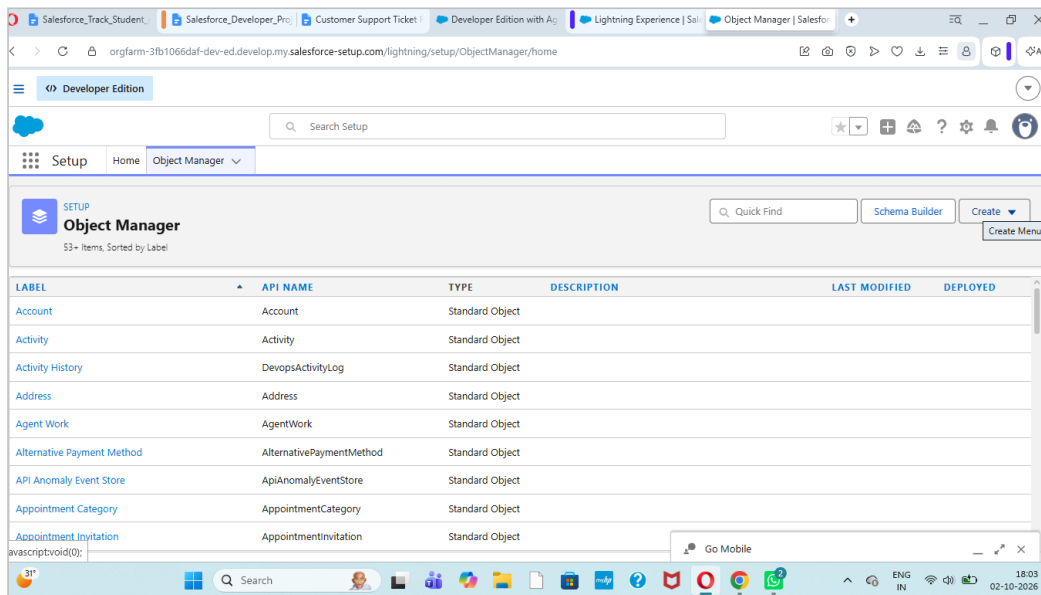


Figure 2: Object Manager in Setup; the Create menu starts a new custom object

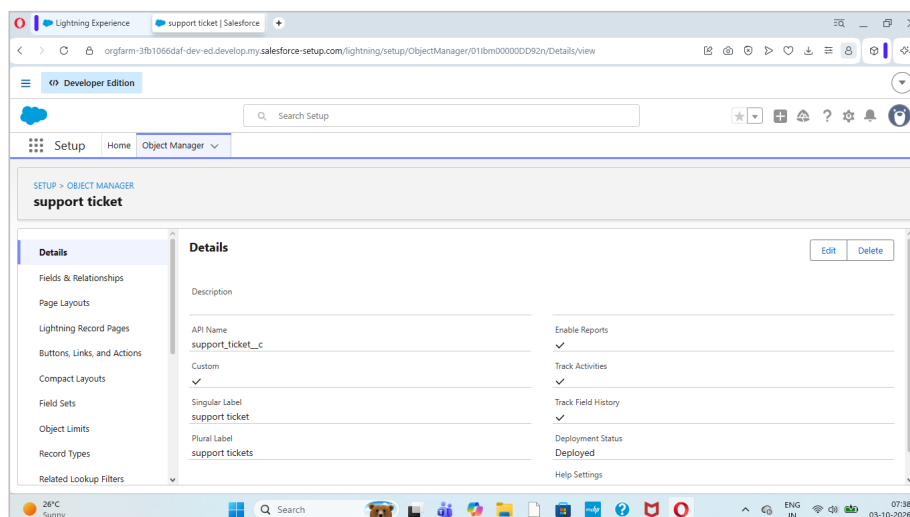


Figure 3: Object details page after saving

Activity 2: Create the fields

Fields are added from Object Manager > support ticket intelligence > Fields and Relationships > New. The field labels below are written exactly as they appear in the org.

Field label	API name	Data type	Values / description
ticket number	Name	Auto Number	TKT-{0000}, record name
subject	subject__c	Text (255)	Short title of the problem. Required.
description	description__c	Long Text Area (32000)	Full issue details. Required.
category	category__c	Picklist	Billing, Technical Issue, Account Access, Product Inquiry, Refund, Other
customer type	customer_type__c	Picklist	Free, Paid, Premium
channel	channel__c	Picklist	Email, Phone, Chat, Social Media
status	status__c	Picklist	Open, In Progress, Resolved, Closed
priority	priority__c	Picklist	Low, Medium, High, Critical (priority entered by the user)
predict priority	predict_priority__c	Picklist	Low, Medium, High, Critical (filled by the Flow)
priority score	priority_score__c	Number (3,0)	Filled by the Flow
assigned team	assigned_team__c	Text (100)	Filled by the Flow
customer	customer__c	Lookup (Account)	Account the ticket belongs to

For each picklist: choose Picklist as the data type, enter the label, type one value per line, then select Next, Next and Save. Subject and description are marked as required. The three Flow-filled fields are not required.

Activity 3: Link tickets to Accounts and add sample data

1. Create the Customer lookup field: New > Lookup Relationship > Related To: Account > Label: Customer > Save.
2. Create a test Account (App Launcher > Accounts > New). The test account used in this project is named **test customer**.
3. Open the support ticket intelligence tab and create sample tickets with different subjects, categories and customer types.
4. Edit one ticket so that Customer is the test Account and the description contains the word urgent.

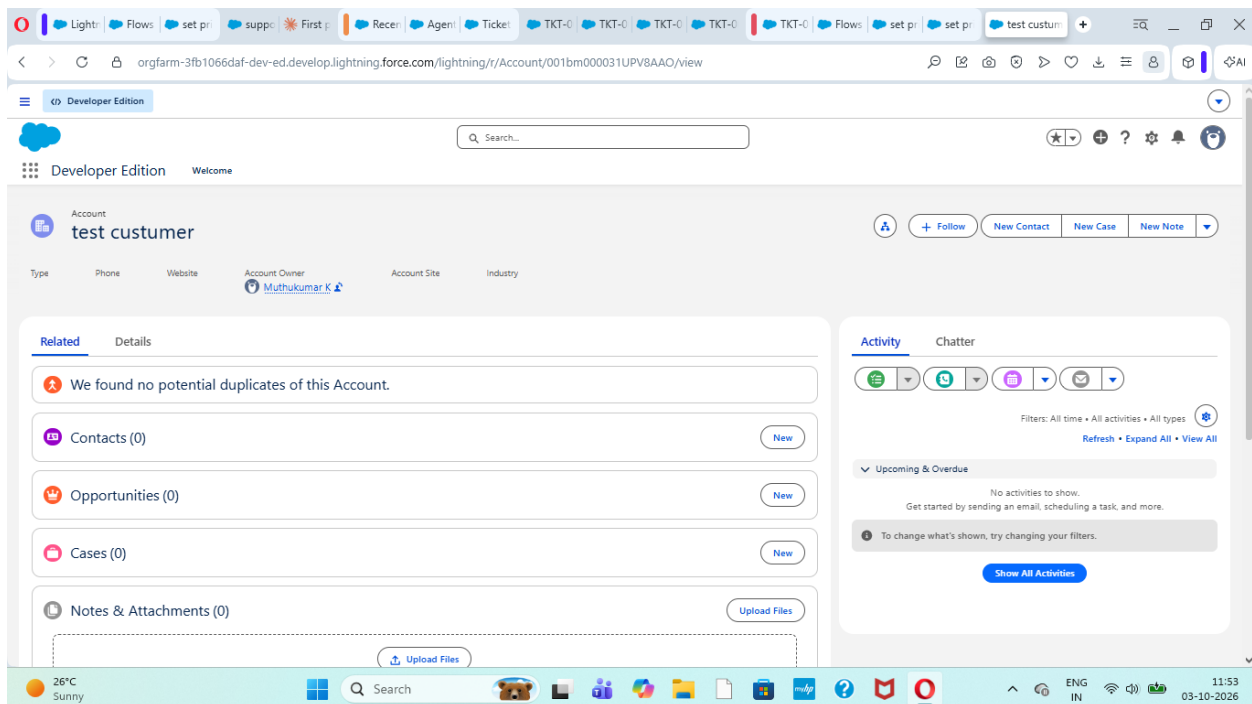
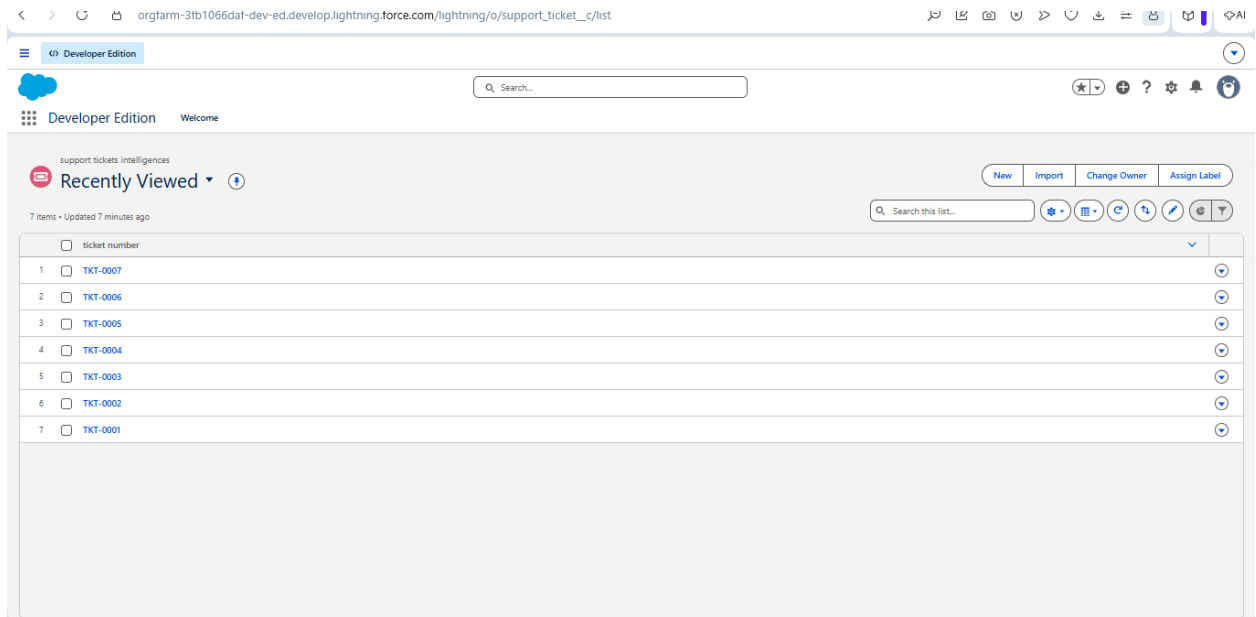


Figure 6: Test Account record (test customer)



orgtarm-3fb1066da1-dev-ed.develop.lightning.force.com/lightning/o/support_ticket_c/list

Developer Edition

support tickets intelligences

Recently Viewed

7 items • Updated 7 minutes ago

Search this list...

	ticket number	
1	TKT-0007	
2	TKT-0006	
3	TKT-0005	
4	TKT-0004	
5	TKT-0003	
6	TKT-0002	
7	TKT-0001	

Figure 7: List view of the seven sample support tickets

Milestone 3: Record-Triggered Flow (set priority)

This Flow runs when a ticket is created. It reads the ticket fields and writes the predicted priority, the score and the support team, so no one has to do it by hand.

Activity 1: Create the Flow

1. Setup > Flows > New Flow > Record-Triggered Flow > Create.
2. Object: support ticket intelligence. Trigger: A record is created. Optimize the flow for Fast Field Updates.
3. Save the Flow with the label **set priority**.

Activity 2: Create three formula resources

Toolbox > New Resource > Resource Type: Formula. The formulas use the Subject field because a Long Text Area field cannot be used inside a formula.

Formula PriorityF (Text):

```
IF(OR(CONTAINS(LOWER({!$Record.subject__c}),"urgent"),
CONTAINS(LOWER({!$Record.subject__c}),"crash"),
CONTAINS(LOWER({!$Record.subject__c}),"down"),
CONTAINS(LOWER({!$Record.subject__c}),"payment failed")), "Critical",
IF(OR(ISPICKVAL({!$Record.customer_type__c},"Premium"),
ISPICKVAL({!$Record.category__c},"Billing"),
ISPICKVAL({!$Record.category__c},"Refund")), "High",
IF(OR(ISPICKVAL({!$Record.category__c},"Technical Issue"),
ISPICKVAL({!$Record.category__c},"Account Access")), "Medium", "Low")))
```

Formula ScoreF (Number, 0 decimal places):

```
IF({!PriorityF}="Critical",90,IF({!PriorityF}="High",70,
IF({!PriorityF}="Medium",50,20)))
```

Formula TeamF (Text):

```
IF(OR(ISPICKVAL({!$Record.category__c},"Billing"),
ISPICKVAL({!$Record.category__c},"Refund")), "Billing Support",
IF(ISPICKVAL({!$Record.category__c},"Account Access"),
"Account Support", "Technical Support"))
```

Activity 3: Add the Assignment element

Between Start and End add an Assignment with the label Set Values and three rows:

Variable	Operator	Value
{!\$Record.predict_priority__c}	Equals	{!PriorityF}
{!\$Record.priority_score__c}	Equals	{!ScoreF}
{!\$Record.assigned_team__c}	Equals	{!TeamF}

Select Done, then Save and Activate the Flow. The decision rules are summarised below.

Output	Rule
Predict priority = Critical	Subject contains urgent, crash, down or payment failed
Predict priority = High	Customer type is Premium, or category is Billing or Refund
Predict priority = Medium	Category is Technical Issue or Account Access
Predict priority = Low	All other tickets
Priority score	Critical 90, High 70, Medium 50, Low 20
Assigned team	Billing or Refund: Billing Support. Account Access: Account Support. Others: Technical Support

The screenshot displays the Salesforce Flow Builder interface for a Record-Triggered Flow named "set priority - V2". The flow is configured with the following elements:

- Start:** Record-Triggered Flow Start.
- Assignment:** An Assignment element with the label "set values" and API Name "set_values". It contains three rows of variable assignments:
 - Variable: `{!$Record.predict_priority__c}`, Operator: `Equals`, Value: `{!PriorityF}`.
 - Variable: `{!$Record.priority_score__c}`, Operator: `Equals`, Value: `{!ScoreF}`.
 - Variable: `{!$Record.assigned_team__c}`, Operator: `Equals`, Value: `{!TeamF}`.
- End:** End node.

The interface also shows a sidebar with flow elements, a top navigation bar, and a bottom status bar.

Figure 9: Record-triggered Flow set priority with the Assignment element open

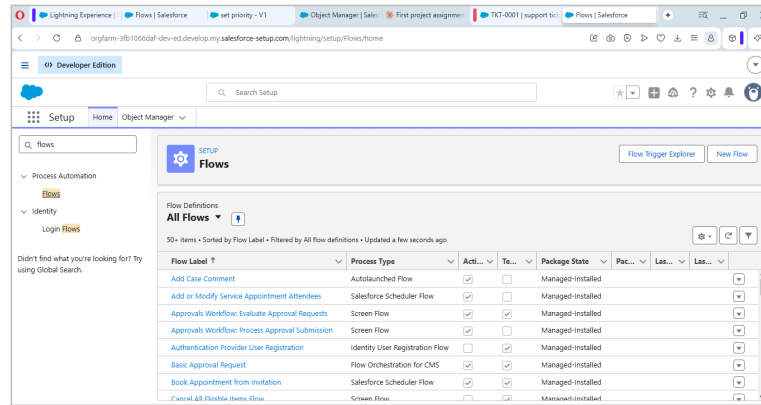


Figure 10: Flows page in Setup (Process Automation > Flows)

Milestone 4: Autolaunched Flow (Support Ticket Priority Flow)

An Agentforce action can only call a Flow that has no trigger, so a second Flow was built for the agent. It takes an Account Name as input, finds the ticket linked to that account, reads its description and returns the priority.

Activity 1: Create the Flow and its variables

1. Setup > Flows > New Flow > Autolaunched Flow (No Trigger) > Create. Save it with the label **Support Ticket Priority Flow**.
2. Create variable **AccountNameInput**: Data Type Text, tick Available for input.
3. Create variables **RecordID**, **PriorityLevel** and **FinalMessage**: Data Type Text, tick Available for output.

Label	API name	Type	Input / Output
Account name	AccountNameInput	Text	Input
Record ID	RecordID	Text	Output
Priority level	PriorityLevel	Text	Output
Final message	FinalMessage	Text	Output

Activity 2: Add the elements**Step 1: Get Account**

Get Records element. Label GetAccount. Object: Account. Condition: Name Equals {!AccountNameInput}. Store only the first record.

Step 2: Get Ticket

Get Records element. Label GetTicket. Object: support ticket intelligence. Condition: customer Equals {!GetAccount.Id}. Store only the first record.

Step 3: Decision (Analyze Description)

Outcome	Condition (description contains)
High Priority	urgent, not working, failure (any condition is met)
Medium Priority	slow, delay (any condition is met)
Default outcome: Low Priority	none of the above words

Step 4: Assignments (one for each path)

Path	RecordID	PriorityLevel	FinalMessage
Set High	{!GetTicket.Id}	High	High priority ticket, dedicated task created and assigned to senior support agent.
Set Medium	{!GetTicket.Id}	Medium	Medium priority ticket, scheduled for standard support review.
Set Low	{!GetTicket.Id}	Low	Low priority ticket, handled in normal queue.

Finally Save the Flow and select Activate. The final structure is: Start > Get Account > Get Ticket > Decision (Analyze Description) > Assignment for the chosen priority > End.

Activity 3: Test the Flow with Debug

Open the Flow and select Debug. Enter the Account Name of the test account and select Run. The debug log shows each element that ran.

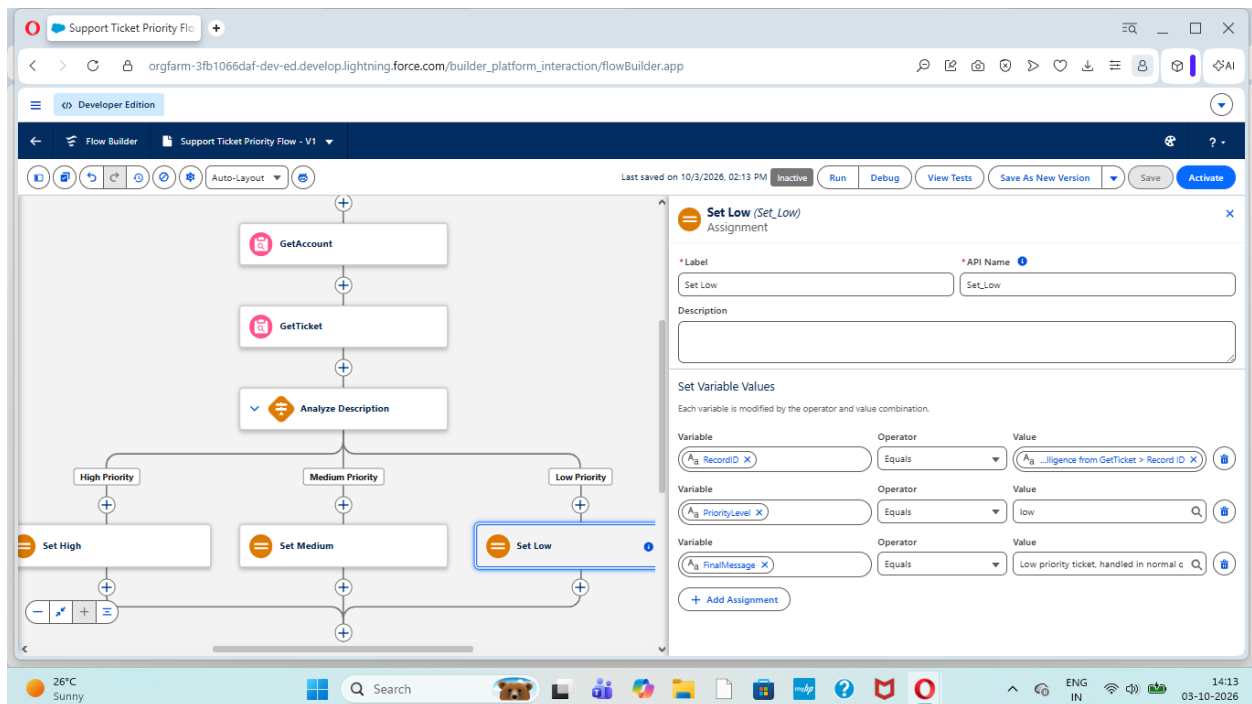


Figure 12: Autolaunched Flow canvas: GetAccount, GetTicket, Analyze Description and the Set High, Set Medium and Set Low assignments

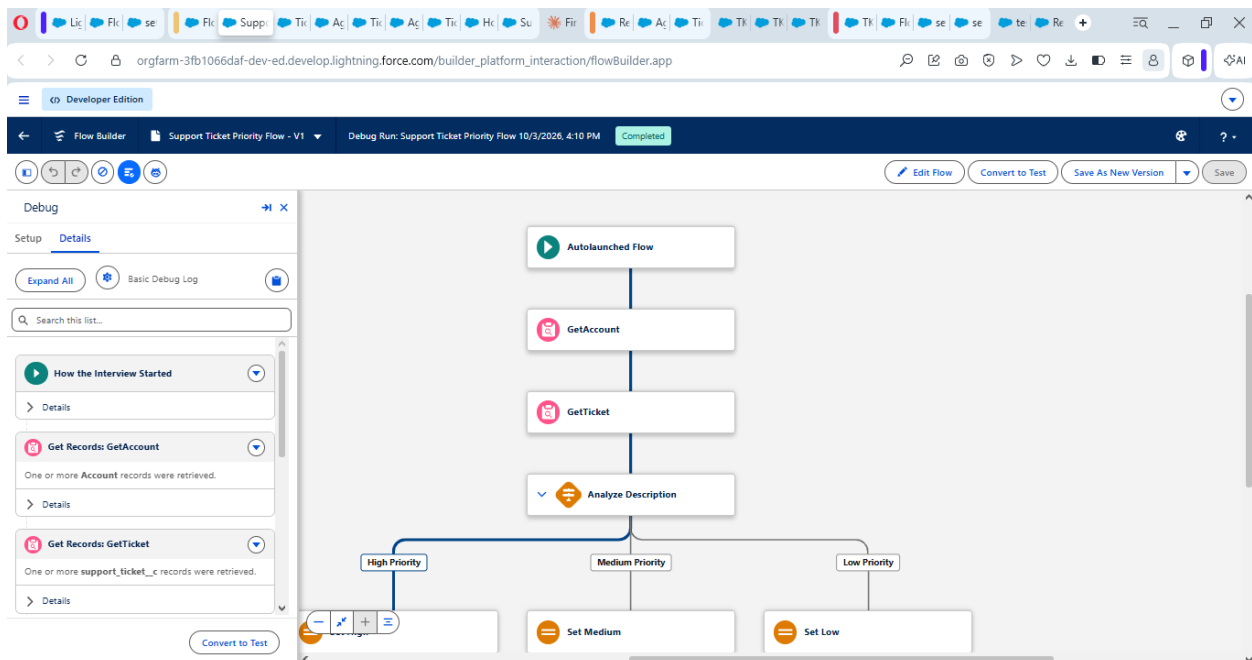


Figure 13: Flow Debug run completed, with the debug log on the left and the flow path highlighted

Milestone 5: Agentforce Configuration

Activity 1: Create the agent

1. Setup > Agentforce Agents (turn Agentforce on if asked).
2. Select New Agent and choose an Agentforce Employee Agent. Name it **Ticket Triage Agent** and add a short description.
3. Select Create. The agent opens in Agentforce Builder.

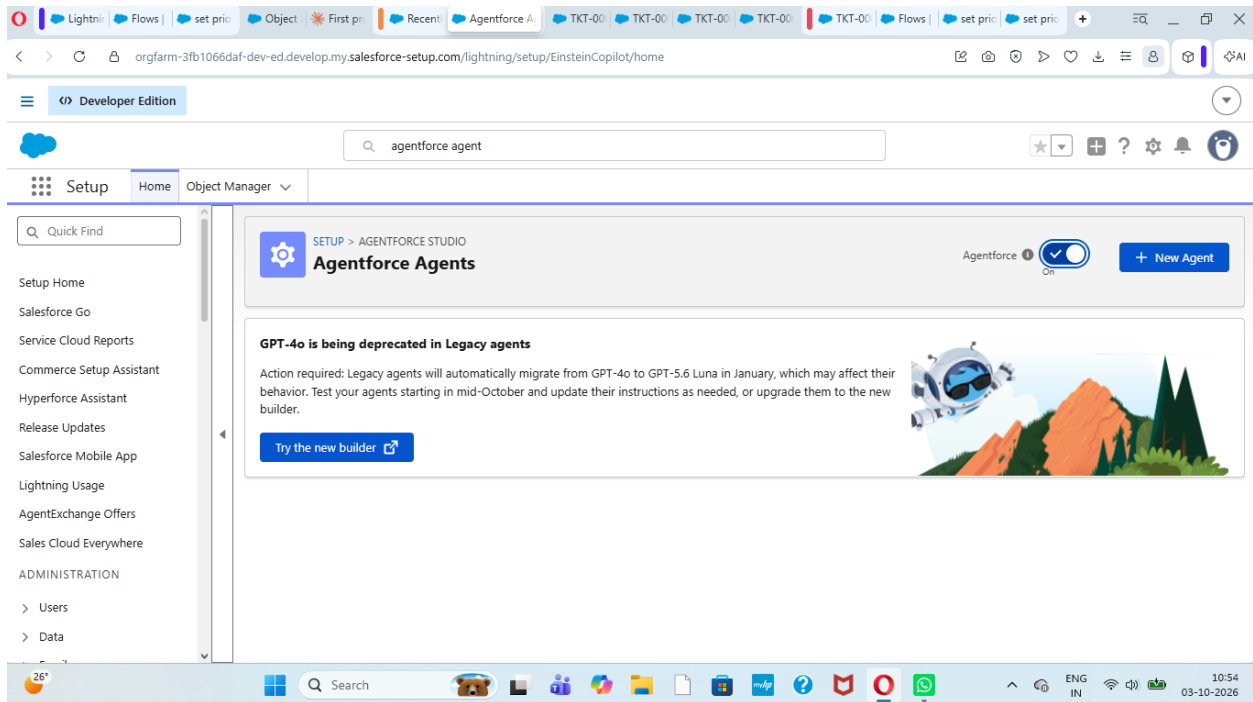


Figure 14: Agentforce Agents page in Setup (Agentforce turned on)

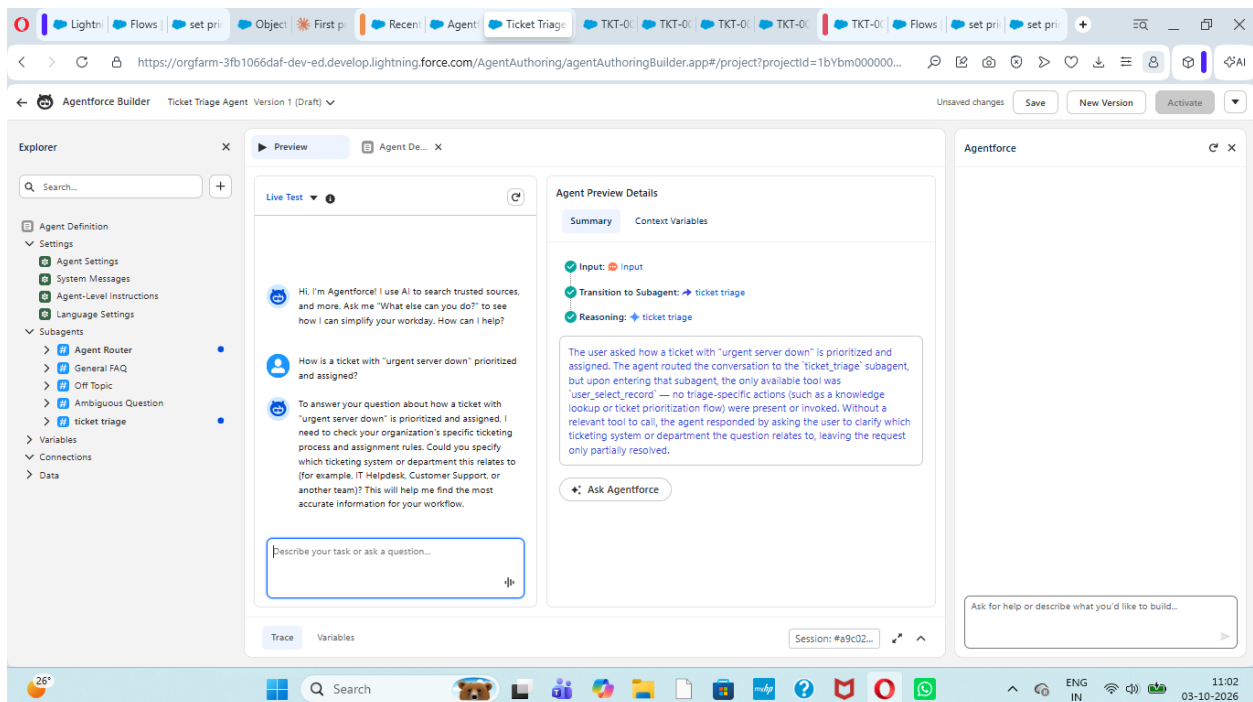


Figure 15: Agentforce Builder: Explorer with the agent and its subagents, and the Preview panel

Activity 2: Create the subagent

In Agentforce Builder create a new subagent with the following configuration.

Item	Value
Name	Support Ticket Priority Analysis
API name	Support_Ticket_Priority_Analysis
Classification description	Analyzes customer support tickets to determine priority based on the issue description and urgency. Handles user requests about ticket priority.
Scope	Only analyzes support ticket descriptions and determines the priority level (High, Medium, Low). Does not handle unrelated requests such as billing, subscription management or account updates.

Instructions

Ask the user for the account name. Then run the flow action with that account name. Show the Record ID, Priority Level and Final Message from the result. Priority keywords are: High for urgent, not working and failure; Medium for slow and delay; Low when none of these words are found.

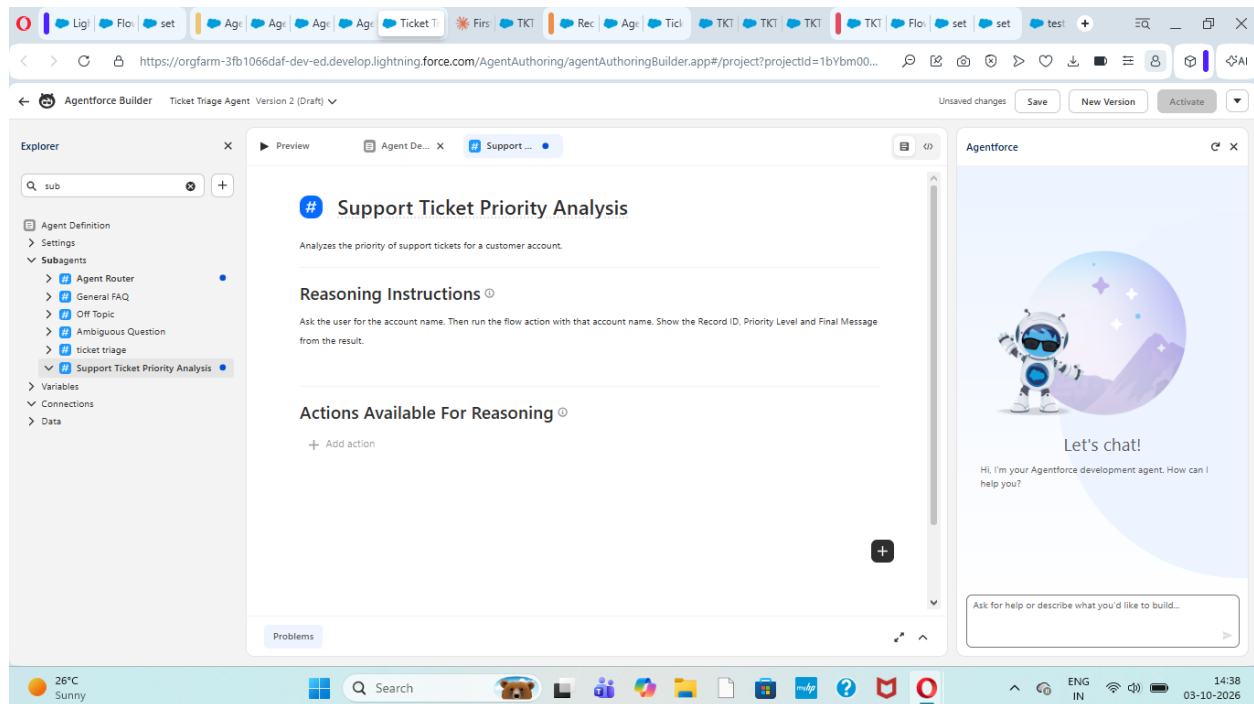


Figure 16: Subagent Support Ticket Priority Analysis with its description and reasoning instructions

Activity 3: Create the agent action

The Autolaunched Flow is exposed to the agent as an agent action. Setup > Agentforce Assets > Actions > New Agent Action.

Setting	Value
Reference Action Type	Flow
Reference Action	Support Ticket Priority Flow
Agent action description	Finds the support ticket for a customer account name and returns its record ID, priority level and a final message.
Loading text	Checking your ticket priority...
Input: AccountNameInput	Description: Name of the customer account whose ticket priority is needed. Require input and Collect data from user are enabled.
Input: accountid	An extra numeric input of the Flow that the logic does not use. Its description tells the agent not to ask the user for it.
Outputs: RecordID, PriorityLevel, FinalMessage	Each has a description and Show in conversation is enabled.

The action is then assigned to the subagent so that the agent can run it during a conversation. In Agentforce Builder this is done from the subagent screen: under Actions Available For Reasoning select Add action (or Select action), then either choose the action from the Asset Library or choose Create a custom action, set Reference Action Type to Flow and pick the Flow [4][5]. The subagent's reasoning instructions should name the action so that the agent knows when to run it.

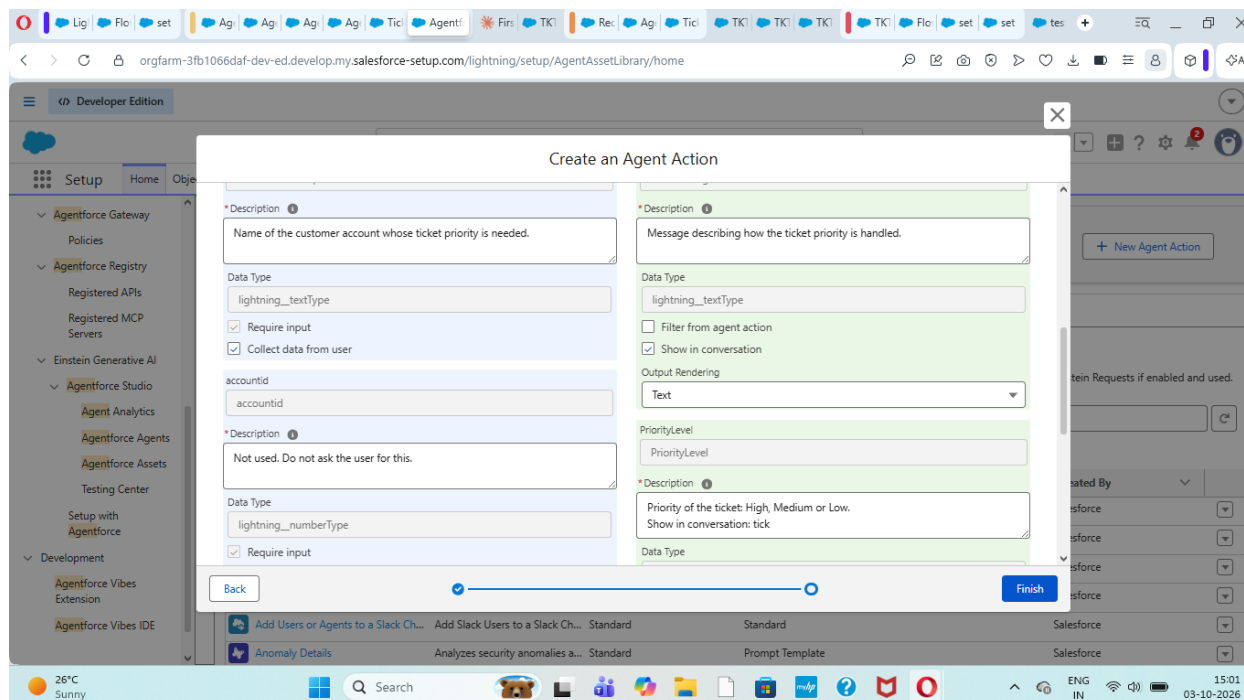


Figure 17: Create an Agent Action wizard: input and output descriptions

Activity 4: Test in the Agent Preview

Open the Preview panel in Agentforce Builder and ask a question about ticket priority. In this test the question "How is a ticket with urgent server down prioritized and assigned?" was routed to the ticket triage subagent. That subagent had no ticket priority action yet, so the agent asked a clarifying question instead of returning a result. This is why the Support Ticket Priority Analysis subagent and its Flow action were set up in the earlier activities. The full result of the Flow is shown by running the same Flow in Flow Debug (Milestone 4, Activity 3).

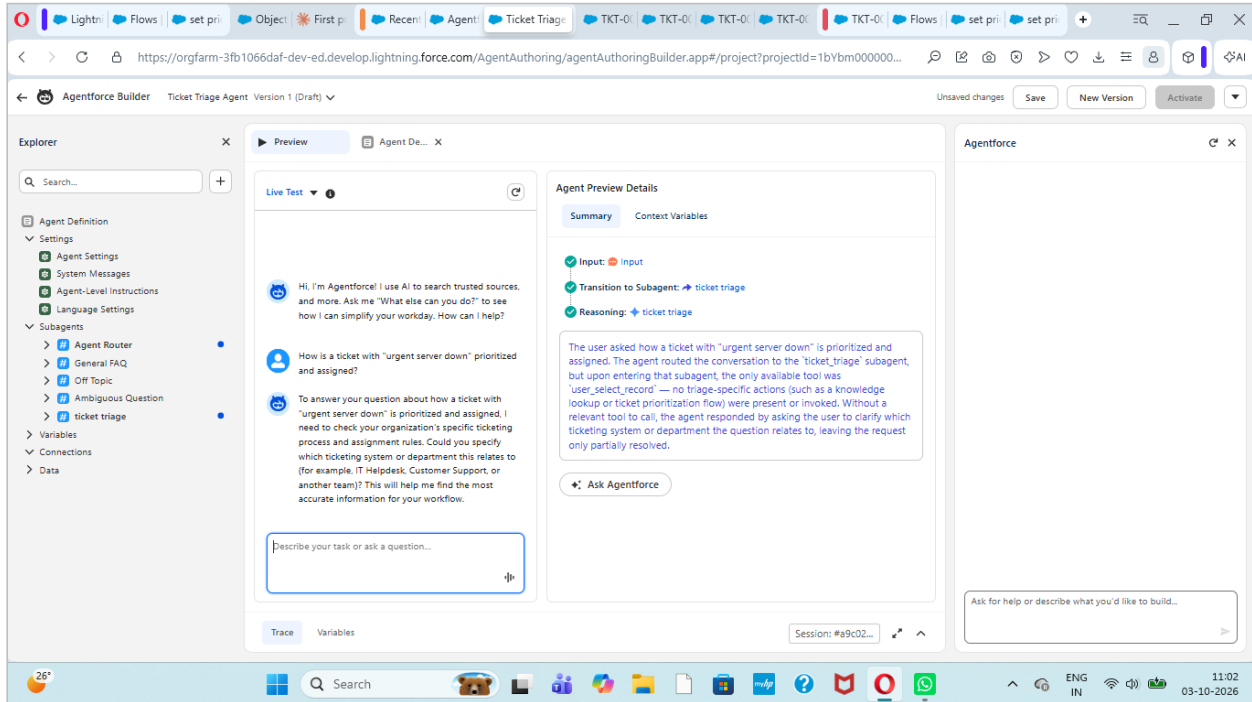


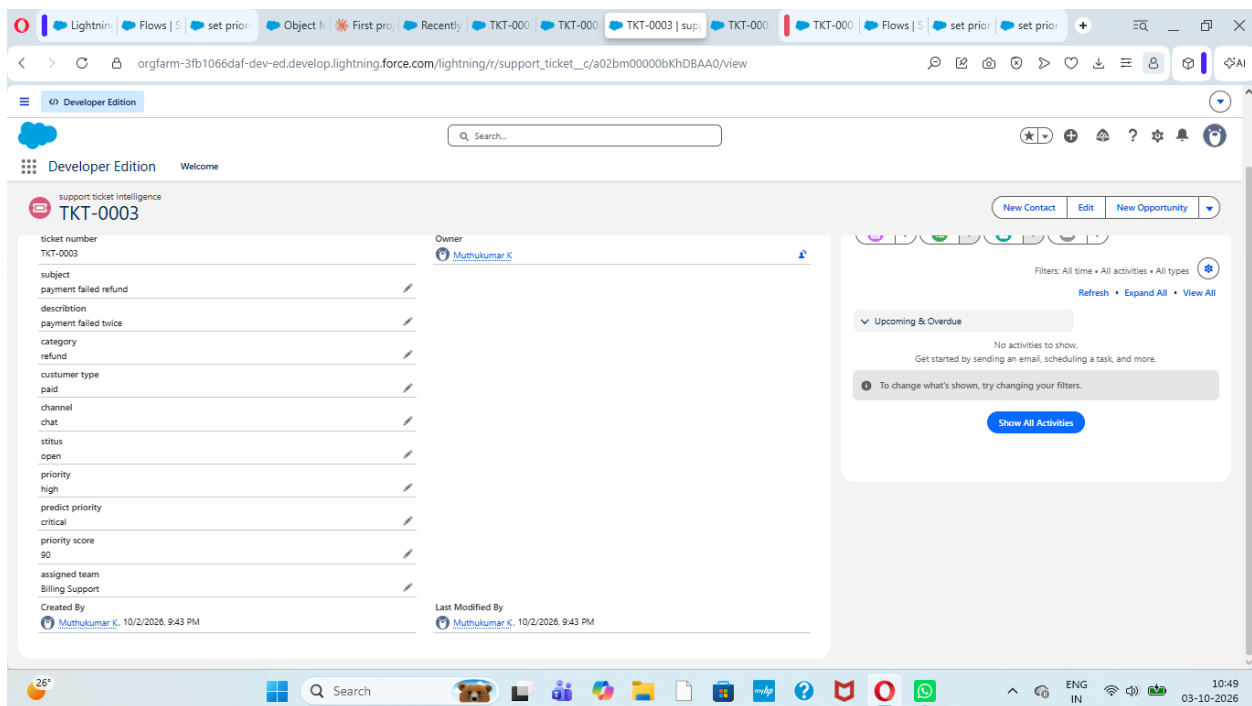
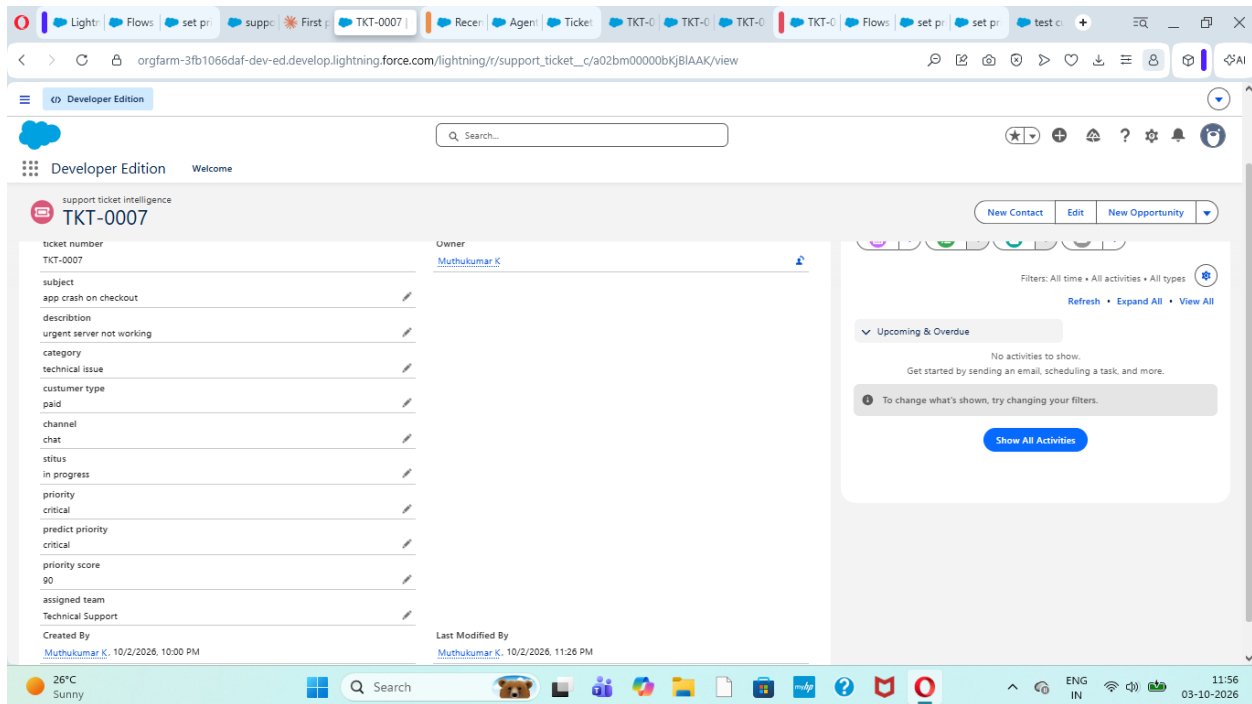
Figure 19: Agent Preview conversation (Ticket Triage Agent, Version 1)

7. FUNCTIONAL AND PERFORMANCE TESTING

7.1 Test cases

Test	Input	Expected result	Actual result	Status
TC-1 Record-triggered Flow, billing ticket	Subject: payment failed refund. Category: Refund. Customer type: Paid. Predicted fields left blank (TKT-0003)	Critical, score 90, Billing Support	Predict priority Critical, score 90, Assigned team Billing Support	Pass
TC-2 Record-triggered Flow, technical ticket	Subject: app crash on checkout. Category: Technical Issue. Customer type: Paid (TKT-0007)	Critical, score 90, Technical Support	Predict priority critical, score 90, Assigned team Technical Support	Pass
TC-3 Record-triggered Flow, low ticket	Subject: how to change profile. Category: Other. Customer type: Free (TKT-0004)	Low, score 20, Technical Support	Predict priority low, score 20, Assigned team Technical Support	Pass
TC-4 Autolaunched Flow, existing account	Account name of the test account whose ticket description contains urgent	GetAccount and GetTicket find records, High Priority outcome	Debug log: records retrieved, High Priority outcome, Assignment Set High, 3 variables updated	Pass
TC-5 Autolaunched Flow, account not found	Account name that does not exist (wrong spelling)	No account found, default Low outcome	Debug log: no records retrieved for GetAccount, default outcome, Assignment Set Low	Pass (confirms exact name is needed)
TC-6 Agent conversation	Ask how an urgent server down ticket is prioritized and assigned	Agent runs the priority Flow action	Routed to ticket triage subagent, which asked a clarifying question (Figure 19)	Partial (action to be added)

7.2 Test evidence



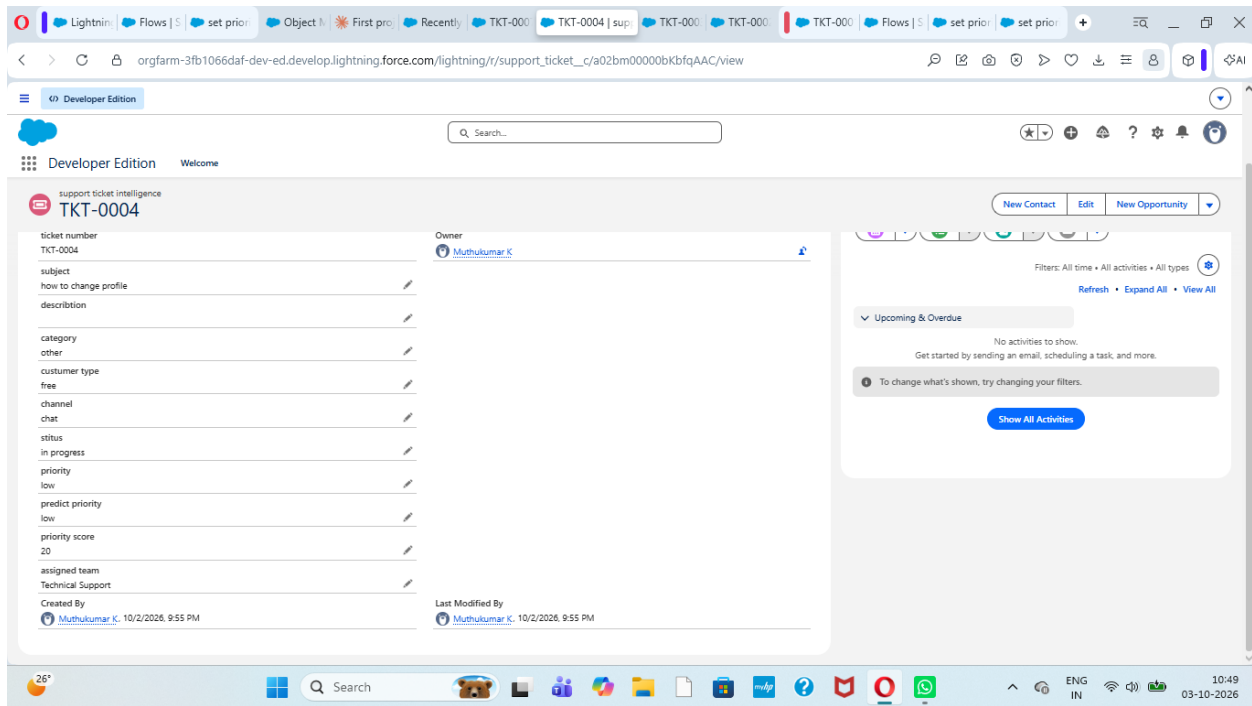


Figure 22: TKT-0004 after saving: Low, score 20, Technical Support

7.3 Performance

The Autolaunched Flow finished in 1.25 seconds in Flow Debug (1.87 seconds for the run in which no account was found). The record-triggered Flow fills the fields at the moment the record is saved, so the user sees the result on the ticket page immediately.

7.4 Observations

- The account name entered into the Flow must match the Account record exactly. A wrong spelling returns no account and the Flow falls to the default Low outcome.
- Formulas cannot read a Long Text Area field, so the record-triggered Flow reads the Subject field for keywords. The Autolaunched Flow reads the description field through a Decision element, which can use Long Text Area fields.
- The Flow runs only when a ticket is created. Tickets created earlier are not changed.

7.5 Issues Found and Fixed

Testing found three issues. They were fixed before the final test cases were recorded.

Issue 1: priority score and assigned team stayed blank

In the first version of the record-triggered Flow, the ticket TKT-0002 received the predicted priority, but the Priority Score and Assigned Team fields stayed empty (Figure below). The Assignment element had two rows with empty values after their variables were changed. The values ScoreF and TeamF were selected again for those two rows, the Flow was saved and activated, and the next ticket (TKT-0003, Figure 21) had all three fields filled.

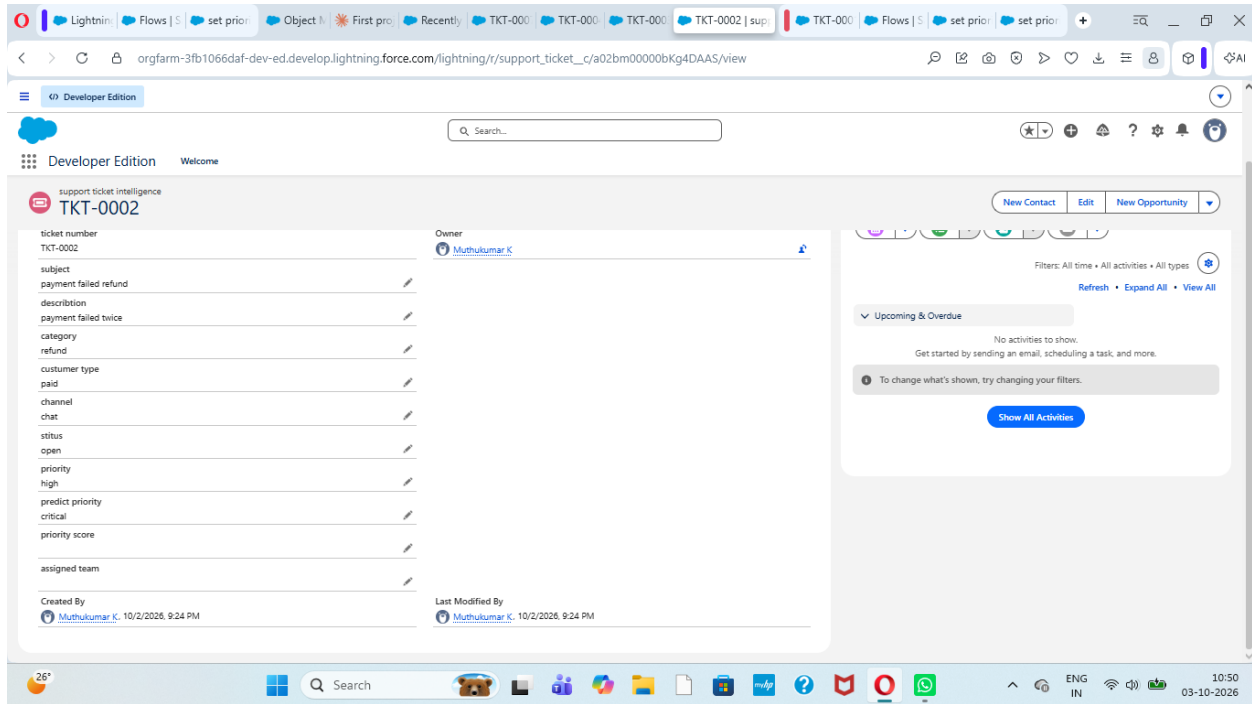


Figure 23: TKT-0002 before the fix: predict priority is filled, priority score and assigned team are blank

Issue 2: a wrongly spelled account name returned Low

The Autolaunched Flow matches the Account Name exactly. When an account name that does not exist was entered, GetAccount found no record and the Decision took the default Low outcome. The Flow was tested again with the exact name of the test account and the High Priority outcome ran. A future version can add a check for an empty result and ask the user to correct the name.

Issue 3: formulas cannot read the Long Text Area field

The description field is a Long Text Area, which cannot be used inside a formula. The record-triggered Flow therefore reads the Subject field, while the Autolaunched Flow reads description through a Decision element, which supports Long Text Area fields.

8. ADVANTAGES AND DISADVANTAGES

Advantages

- **Automatic prioritization:** a new ticket gets a predicted priority the moment it is saved.
- **Priority score:** a numeric score allows tickets to be sorted and filtered.
- **Automatic team assignment:** the Assigned Team field is filled from the ticket category.
- **Faster identification of critical issues:** keywords such as urgent, crash, down and payment failed raise the priority.
- **Conversational access:** a user can ask the Agentforce agent for a customer's ticket priority.
- **No code:** everything is built with clicks (objects, fields, Flow and Agentforce Builder), so it is easy to change.
- **Reduced manual effort:** support staff no longer sort every ticket by hand.
- **Scalable:** new categories, keywords and teams can be added by editing the Flow.

Disadvantages

- **Rule based logic:** the priority depends on configured keywords and field values. It is not a machine learning model and may miss urgency expressed in other words.
- **Limited context understanding:** the rules do not read the full meaning of a ticket.
- **Data dependency:** a ticket must be linked to the right Account for the lookup Flow to find it, and the account name must be spelled exactly.
- **Team stored as text:** the Assigned Team is a text value, not a queue or a user, so it does not route ownership automatically.
- **Agent dependency:** agent responses depend on correct subagent instructions and action configuration, and they may need tuning.
- **Created records only:** the record-triggered Flow works on new tickets and does not re-evaluate existing ones.
- **Maintenance:** keywords, scores and team rules must be updated when support needs change.

9. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System shows how Salesforce Flow and Agentforce can reduce manual ticket handling. A new ticket is given a predicted priority, a priority score and a support team as soon as it is saved. A second Flow finds the ticket linked to an Account and classifies it, and an Agentforce subagent with a flow-based action is configured to start that Flow from a conversation. The solution supports faster handling of urgent tickets, more consistent routing and less repetitive work for the support team. The work was completed in a Salesforce Developer Edition org using only configuration tools.

10. FUTURE SCOPE

- Create a Task for the senior support agent automatically when a ticket is High priority.
- Create Queues for Billing, Account and Technical Support and set the ticket owner to the matching queue.
- Replace keyword rules with a prompt template or Einstein prediction that reads the full ticket text.
- Add SLA monitoring and escalation for older unresolved tickets.
- Add more variables to the Flow and show them in the Agentforce action for richer answers.
- Add dashboards for priority trends and resolution time.
- Make the Agentforce conversation return the priority in one step for every account name.

11. APPENDIX

A. Custom object

support ticket intelligence (API name support_ticket__c) with the fields listed in Milestone 2.

B. Key Salesforce records

Account (standard object) and the support ticket intelligence tickets.

C. Flows

- set priority: record-triggered Flow, runs when a ticket is created.
- Support Ticket Priority Flow: Autolaunched Flow with input AccountNameInput and outputs RecordID, PriorityLevel and FinalMessage.

D. Agentforce

- Agent: Ticket Triage Agent (Employee Agent).
- Subagent: Support Ticket Priority Analysis.
- Agent action: Support_Ticket_Priority_Flow, a Flow-based action in Agentforce Assets.

E. Source files

The GitHub repository contains this report in PDF form and the screenshots used in each milestone.

12. REFERENCES

The numbers in square brackets in the report refer to these sources.

No.	Source
[1]	Salesforce, The Seventh Edition State of Service Report (2025). https://www.salesforce.com/resources/research-reports/state-of-service/
[2]	Salesforce, Inside the Seventh Edition of the State of Service Report. https://www.salesforce.com/blog/state-of-service/
[3]	Salesforce, AI Expected to Resolve Half of Service Cases by 2027, Data Shows (2025). https://www.salesforce.com/news/stories/state-of-service-report-announcement-2025/
[4]	Trailhead, Add a Flow as an Agent Action (Agent Customization with Flows). https://trailhead.salesforce.com/content/learn/modules/agent-customization-with-flows/add-a-flow-as-an-agent-action
[5]	Trailhead, Build with Agentforce Builder (Quick Start: Assemble a Service Agent with Agentforce Builder). https://trailhead.salesforce.com/content/learn/modules/quick-start-assemble-a-service-agent-with-agentforce-builder/build-with-agentforce-builder
[6]	Salesforce Help, Create a Custom Agent Action in the Legacy Builder. https://help.salesforce.com/s/articleView?language=en_US&id=ai.copilot_actions_custom.htm&type=5
[7]	Salesforce, Introducing the Agentforce Guide to Reasoning, Topics, Instructions and Actions. https://www.salesforce.com/agentforce/guide/
[8]	Salesforce Developers, Use HXL Widgets as Agentforce Action Output (Agentforce actions and subagents). https://developer.salesforce.com/docs/platform/hxl/guide/agentforce-action-output.html

Thank You